

QUANTITATIVE METHODS LEVEL 1

Trustville

Hanoi, Jun 2026

Lời giới thiệu

Tài liệu này được biên soạn và tổng hợp chủ yếu từ nguồn học liệu Curriculum level 1. Bên cạnh những trích dẫn, công thức và biểu đồ, đôi khi bạn sẽ gặp một số **đoạn văn tiếng Việt in nghiêng và màu xanh** là phần diễn giải bổ sung của Giảng viên biên soạn, hy vọng giúp giải thích rõ ràng hơn.

Để môn học trở nên hiệu quả, các bạn học viên nên sử dụng cùng với một số tài liệu bổ trợ dưới đây:

- Spreadsheet mô phỏng một số ví dụ tính toán, Click
- Mindmap tổng hợp các mục kiến thức chính, Click
- Video bài giảng theo chương trình năm 2024, Click
- Tổng hợp các tài liệu khác: Click.

Xin gửi lời cảm ơn trân trọng đến đội ngũ Giảng viên của Trustville, và những bạn học viên đã đóng góp nhiều ý tưởng để giúp hoàn thiện tài liệu này.

Mặc dù tài liệu biên soạn được chuẩn bị một cách kỹ lưỡng và cẩn trọng, nhưng cũng không thể tránh khỏi sai sót và lỗi diễn đạt. Vậy rất mong những ý kiến phản hồi từ các bạn khi sử dụng tài liệu, để giúp hoàn thiện nội dung hơn.

Mọi ý kiến đóng góp xin gửi về email: dongkhautu@gmail.com.

Chúc các bạn có những giờ học bổ ích và hoàn thành mục tiêu!

Table of Contents

1	Returns of Financial Assets and Instruments	5
1.1	FINANCIAL RETURNS	5
1.2	COMMON RETURN MEASURES	7
2	Types of Returns for Financial Assets, Instruments, and Indicators	11
2.1	TYPES OF RETURNS	11
2.2	LONG-TERM FINANCIAL ASSET RETURNS	13
2.3	RISKS AND RETURNS	14
3	Benchmarking Returns	17
4	The Time Value of Money in Finance	19
5	Statistical Characteristics of Asset Returns	21
6	Statistical Distributions for Financial Asset Prices and Returns	23
7	Estimation and Hypothesis Testing	25
8	The Return and Risk of a Financial Portfolio	27
9	Simulation of Financial Asset Prices and Returns	29
10	Applications of Simple Linear Regression in Finance	31
11	Introduction to Financial Data Science	33

Module 1

Returns of Financial Assets and Instruments

LEARNING OUTCOMES

a	describe, compare, and interpret returns
b	describe, compare, and interpret required rates of return, risk-free rates, risk premia, and inflation

Financial Returns

Total Return

Arithmetic and Geometric Holding Period Return

Compounding: Daily, Monthly, Quarterly, and Annual Returns

Continuous Compounding

Common Return Measures

The Required Rate of Return

Excess Returns

Real Returns

Gross and Net Return

Pre-Tax and After-Tax Nominal Return

Leveraged Returns

1.1 FINANCIAL RETURNS

- a | describe, compare, and interpret returns

Lãi suất hay lợi suất là đối tượng thống kê chính trong hoạt động đầu tư. Có thể hiểu, thị trường tài chính là nơi buôn bán một loại hàng hóa đặc biệt, là Vốn. Và lãi suất chính là mức giá cho loại hàng hóa đặc biệt đó. Hãy bắt đầu tìm hiểu cách chúng ta đo lường mức giá này.

1.1.1 Total Return

Total Return = Capital Appreciation Return + Capital Distribution Return

$$\begin{aligned} r &= r_{\text{price}} && + r_{\text{distribution}} \\ &= \frac{P_1 - P_0}{P_0} && + \frac{Inc}{P_0} \end{aligned} \quad (1.1)$$

For capital distributions on equities, we focus on the dividend yield; for bonds, we look at current yield.

- Some expressions of Return:

Expected returns (ex ante returns): are the anticipated returns on an investment over a specified period.

Actual returns (ex post returns): are the returns on an investment after the investment period has ended.

Unrealized returns: changes in value represent while an investor holds an investment.

Realized returns: when the investment is ultimately sold, gains and losses would be realized at the current market value.

1.1.2 Arithmetic and Geometric Holding Period Return

To ensure accurate comparisons across different investments and time periods, it's crucial that these returns are calculated using a uniform time period.

- **arithmetic mean return:**

$$\bar{r}_A = \frac{r_1 + r_2 + \cdots + r_T}{T} = \frac{1}{T} \sum_{t=1}^T r_t \quad (1.2)$$

- **geometric mean return:**

$$\bar{r}_G = \sqrt[T]{(1 + r_1) \times (1 + r_2) \times \cdots \times (1 + r_T)} - 1 = \left[\prod_{t=1}^T (1 + r_t) \right]^{1/T} - 1 \quad (1.3)$$

Geometric mean is a more accurate reflection of investment performance over time. Arithmetic average return calculations assume that the amount invested at the beginning of the period does not change, but the base amount changes due to capital appreciation or depreciation. The arithmetic-geometric mean inequality highlights the impact of volatility on investment returns.

Exam Tip

Remember: - **Arithmetic Mean** is the best estimator of return for a single future period (ex-ante). - **Geometric Mean** represents the actual historical compound rate of return over multiple periods (ex-post). - Unless all period returns are identical, **Arithmetic Mean is always strictly greater than Geometric Mean** due to volatility ($\bar{r}_A > \bar{r}_G$).

1.1.3 Compounding: Daily, Monthly, Quarterly, and Annual Returns

With c is the number of compounding periods in a year,

$$r_{\text{annualized}} = (1 + r_{\text{period}})^c - 1 \quad (1.4)$$

One major limitation of annualizing returns is the implicit assumption that earned returns can be repeated.

Ví dụ thực tế tại Việt Nam: Nếu một ngân hàng thương mại áp dụng lãi suất kỳ hạn 3 tháng là 1.5% (tương ứng r_{period}), khi quy đổi ra lãi suất năm có ghép lãi (annualized return), ta tính được: $r_{\text{annualized}} = (1 + 0.015)^4 - 1 \approx 6.136\%$. Con số này lớn hơn mức 6.0% nếu chỉ nhân đơn thuần kỳ hạn với 4. Điều này phản ánh giá trị thời gian của tiền khi tiền lãi kỳ trước được tái đầu tư để sinh lãi ở kỳ sau.

1.1.4 Continuous Compounding

The more frequently we compound returns, the higher the annual return will be; however, there is a limit. As shown in Exhibit 1.1:

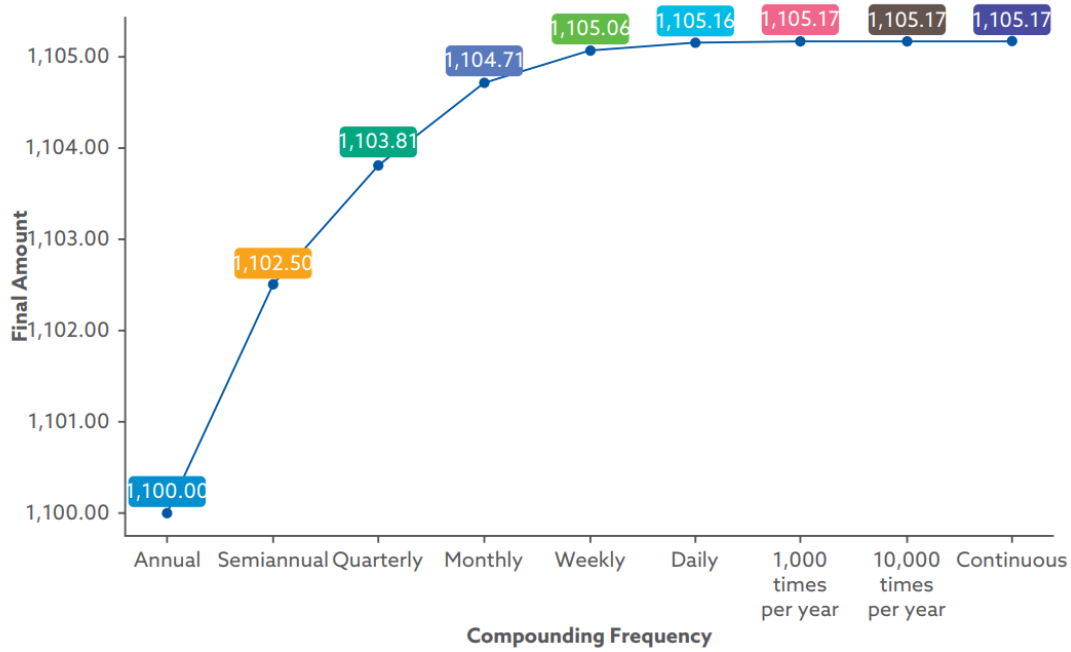


Figure 1.1: The Impact of Compounding

- **continuously compounded return:**

$$\tilde{r}_{t,t+1} = \ln\left(\frac{P_{t+1}}{P_t}\right) = \ln(1 + r_{t,t+1}) \quad (1.5)$$

The additive property of continuously compounded returns provides calculation advantages:

$$\begin{aligned} \frac{P_T}{P_0} &= \frac{P_T}{P_{T-1}} \cdot \frac{P_{T-1}}{P_{T-2}} \cdots \frac{P_2}{P_1} \cdot \frac{P_1}{P_0} \\ \Rightarrow \ln\left(\frac{P_T}{P_0}\right) &= \ln\left(\frac{P_T}{P_{T-1}}\right) + \ln\left(\frac{P_{T-1}}{P_{T-2}}\right) + \cdots + \ln\left(\frac{P_2}{P_1}\right) + \ln\left(\frac{P_1}{P_0}\right) \\ \Rightarrow \tilde{r}_{0,T} &= \tilde{r}_{T-1,T} + \tilde{r}_{T-2,T-1} + \cdots + \tilde{r}_{1,2} + \tilde{r}_{0,1} = \sum_{t=1}^T \tilde{r}_{t-1,t} \end{aligned}$$

Tại sao trong phân tích định lượng nâng cao, người ta lại thích dùng suất sinh lãi ghép lãi liên tục (continuously compounded return) hơn là suất sinh lãi rời rạc thông thường? Đó chính là nhờ tính chất cộng gộp qua thời gian ở trên (time-additivity). Tổng log return của từng ngày chính bằng log return của cả năm, giúp đơn giản hóa các mô hình toán học và thống kê phức tạp.

1.2 COMMON RETURN MEASURES

- b | describe, compare, and interpret required rates of return, risk-free rates, risk premia, and inflation

1.2.1 The Required Rate of Return

- **opportunity cost:** that incurs when delaying consumption, or the value that investors forgo by choosing a course of action.
- **required rate of return:** the minimum rate of return an investor must receive to accept the investment.

Components of Required Rate of Return:

- **risk-free rate:** is the return on an investment without any default or reinvestment risk.

It refers to returns on short-term government bonds, such as the 30-day US Treasury bill. The bonds have a short maturity, which reduces their sensitivity to interest rates changes.

real risk-free rate reflects the time preferences of individuals for current versus future real consumption.

$$r_{\text{real_rf}} = \frac{1 + r_{\text{rf}}}{1 + \text{infl}} - 1 \quad \text{or quick approximation: } r_{\text{real_rf}} \approx r_{\text{rf}} - \text{infl} \quad (1.6)$$

- Total required returns for any asset can then be approximated as:

$$r = r_{\text{rf}} + \text{infl} + \sum_{j=1}^n (f_j \times \text{rp}_j) \quad (1.7)$$

n is the number of risk factors

f_j is the exposure of the security to the j^{th} risk factor

rp_j is the specific risk premium associated with the j^{th} risk factor

$\sum_{j=1}^n (f_j \times \text{rp}_j)$ is the sum of all risk premia.

As shown in Exhibit 1.2:

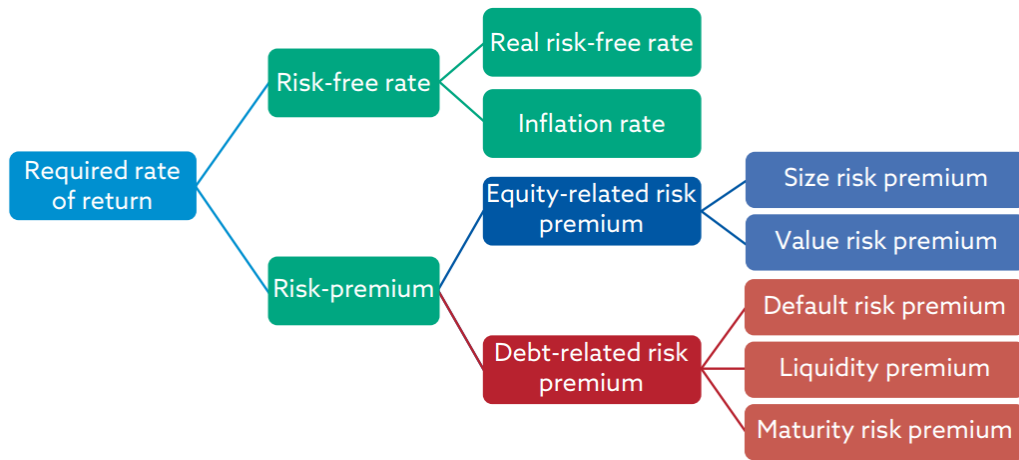


Figure 1.2: Components of Required Rate of Return and Examples of Different Risk Premia

Examples of risk premia for bonds:

- **Default risk premium:** the additional return for buying a riskier corporate bond instead of a less-risky government bond.
- **Liquidity premium:** the additional return for holding a bond that may be harder to sell quickly.
- **Maturity risk premium:** the additional return for holding more interest rate sensitive, longer maturity bonds.

Examples of risk premia for stocks:

- **Size risk premium:** the return spread between small and large market-capitalization stocks.
- **Value risk premium:** the return spread between value stocks (high book-to-market) and growth stocks (low book-to-market).

1.2.2 Excess Returns

Excess returns: the difference between an asset's return and the return on some reference rate or benchmark.

$$\dot{r}_i = \frac{1 + r_i}{1 + r_B} - 1 \quad \text{or quick approximation: } \dot{r}_i \approx r_i - r_B \quad (1.8)$$

1.2.3 Real Returns

Real returns are inflation-adjusted returns.

$$r_{\text{real}} = \frac{1 + r_i}{1 + \text{infl}} - 1 \quad \text{or quick approximation: } r_{\text{real}} \approx r_i - \text{infl} \quad (1.9)$$

1.2.4 Gross and Net Return

Gross return = Total return - Transaction costs

An appropriate measure for evaluating and comparing the investment skill of managers.

Excludes indirect expenses related to the management and administration of an investment.

Net return = Gross return - Management fees - Administrative expenses

The return the investor actually earns.

Exhibit 1.3 depicts the relationship and shows how costs erode investment returns:

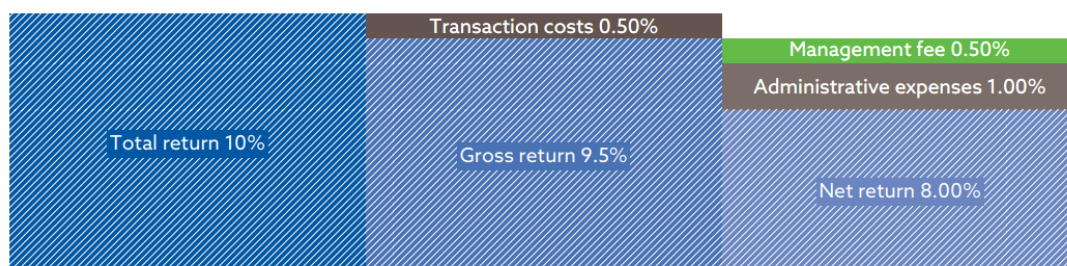


Figure 1.3: The Relationship between Total Return, Gross Return, and Net Return

1.2.5 Pre-Tax and After-Tax Nominal Return

After-tax nominal return:

$$r_{\text{net-tax}} = r_{\text{price}} \times (1 - \text{tax}_{\text{capital gains}}) + r_{\text{capital}} \times (1 - \text{tax}_{\text{capital distribution}}) \quad (1.10)$$

Capital gains tax is a tax on a positive price return. Two forms of capital gains tax rate:

Short-term capital gains are levied on capital gains on short-term holdings.

Long-term capital gains are levied on longer-term holdings and receive preferential tax treatment in a number of countries.

Forms of capital distribution tax rate:

Interest income is taxed as ordinary income in most countries.

Dividend income may be taxed as ordinary income, at a lower rate, or be exempt from taxes depending on the jurisdiction.

Exam Tip

In examinations, note the different tax rates applied to price returns (capital gains tax) versus income distributions (dividend or interest income tax). Always apply them to their respective components of the return formula.

1.2.6 Leveraged Returns

Using leverage to increase the total amount of investible assets is a common feature of investments, whether through borrowing or derivatives. Leverage increases investible capital, magnifying both returns and risks.

$$R_L = R_A + \frac{D}{E}(R_A - R_D) \quad (1.11)$$

R_L is the leveraged return (Return on Equity)

R_A is the return on total assets (unleveraged return)

R_D is the cost of debt (borrowing interest rate)

D/E is the debt-to-equity ratio.

Hãy chú ý: Đòn bẩy tài chính là con dao hai lưỡi. Khi hiệu quả đầu tư trên tài sản (R_A) lớn hơn chi phí đi vay (R_D), đòn bẩy sẽ phóng đại suất sinh lời trên vốn tự có (R_L). Nhưng nếu $R_A < R_D$, điều gì sẽ xảy ra với tỷ suất sinh lời của bạn? Nó sẽ sụt giảm nhanh hơn rất nhiều so với trường hợp không sử dụng đòn bẩy!

The potential financial outcomes for owning a stock outright versus owning a call option on the stock (another form of leverage) are illustrated in Exhibit 1.4:

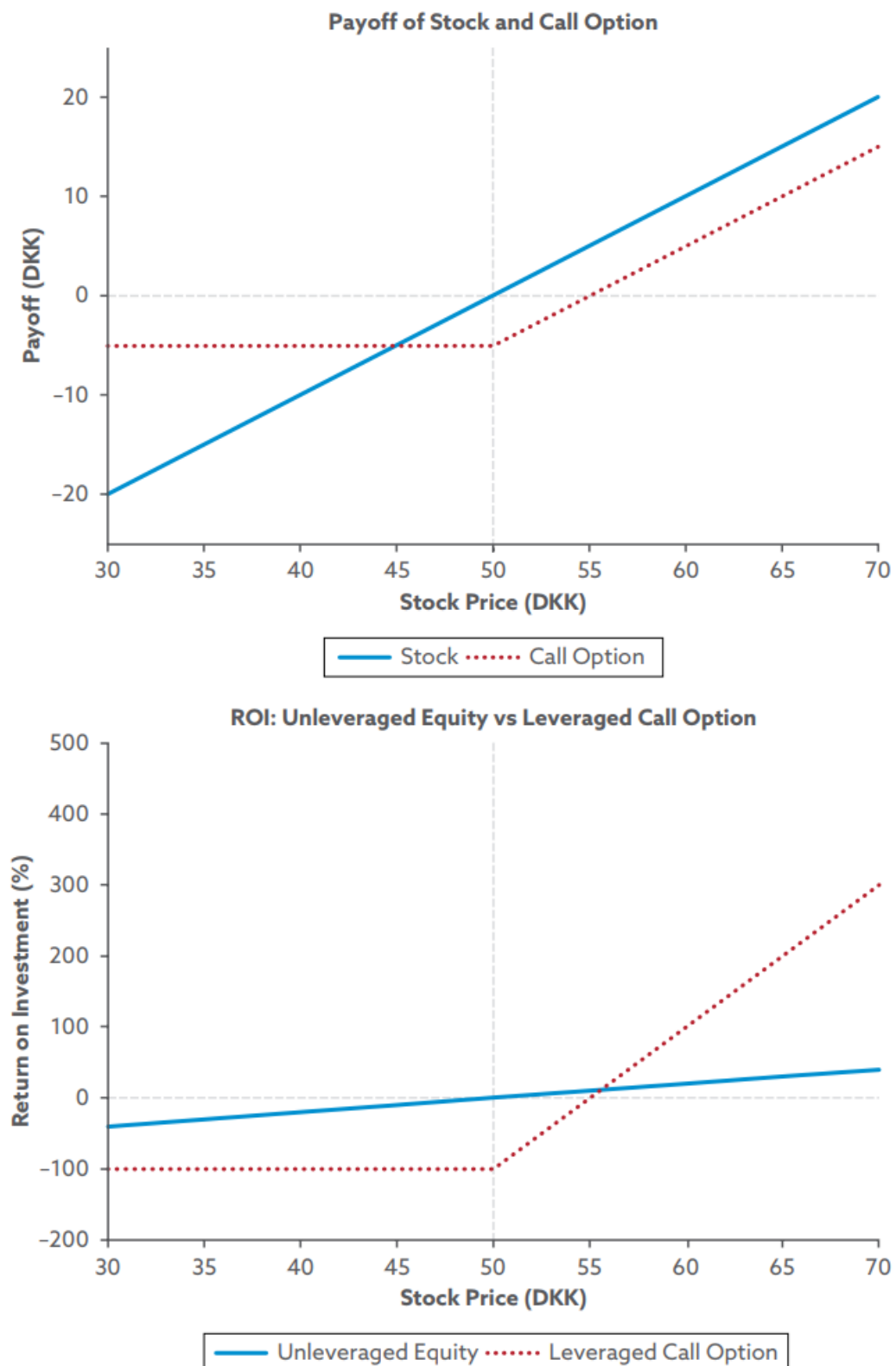


Figure 1.4: The Payoff Relationship between Stock Price, Unleveraged Return, and Leveraged Return

Module 2

Types of Returns for Financial Assets, Instruments, and Indicators

LEARNING OUTCOMES

a	calculate, compare, and interpret different types of returns for financial assets, instruments, and indicators
---	--

Types of Returns

Types of Returns: Financial Assets

Types of Returns: Financial Instruments

Types of Returns: Financial Indicators

Long-Term Financial Asset Returns

Historical Long-Term Asset Returns

Long-Term Asset Return Components

Risks and Returns

Systematic and Unsystematic Risk

The Risk–Return Trade-Off

Portfolio Return Calculation

2.1 TYPES OF RETURNS

a	calculate, compare, and interpret different types of returns for financial assets, instruments, and indicators
---	--

Tỷ suất sinh lời (Return) là thước đo cốt lõi của bất kỳ khoản đầu tư nào. Suất sinh lời tổng thể (Total Return) luôn bao gồm hai thành phần chính: biến động giá (Price Return hay lãi/lỗ vốn) và dòng tiền được phân phối trực tiếp từ tài sản đó (Capital Distribution Return hay cổ tức, tiền lãi). Sự kết hợp của hai cấu phần này thay đổi đáng kể tùy thuộc vào việc chúng ta đang đánh giá tài sản tài chính (Financial Asset), công cụ tài chính (Financial Instrument), hay các chỉ báo tài chính (Financial Indicator). Hãy cùng tìm hiểu cách phân biệt và tính toán chi tiết từng loại!

2.1.1 Types of Returns: Financial Assets

The main financial assets are cash, equity, and debt, as well as hybrids that combine their characteristics.

- **Price Return (Capital Appreciation Return):** reflects changes in a financial asset's price, independent of capital income distributions.

$$r_{\text{price}} = \frac{P_t - P_{t-1}}{P_{t-1}} \quad (2.1)$$

- **Capital Distribution Return (Income Return):** reflects the cash flows received by investors directly from the asset relative to its initial price. This return can be either fixed or variable.

$$r_{\text{distribution}} = \frac{\text{Inc}_t}{P_{t-1}} \quad (2.2)$$

- **Total Return:** the sum of price return and capital distribution return.

$$r_{\text{total}} = r_{\text{price}} + r_{\text{distribution}} = \frac{P_t - P_{t-1} + \text{Inc}_t}{P_{t-1}} \quad (2.3)$$

Lưu ý về tính đồng thời: Khi một tài sản tài chính chi trả một khoản dòng tiền X tại thời điểm t (ví dụ cổ tức), trên lý thuyết, giá của tài sản đó sẽ giảm ngay lập tức một lượng đúng bằng X . Do đó, trong khoảng khắc ngắn hạn đó, phần suất sinh lời phân phối sẽ bù trừ chính xác với phần sụt giảm giá (Price Return). Sau thời điểm đó, tài sản mới tiếp tục chu kỳ biến động giá mới. Bạn đã bao giờ quan sát hiện tượng giá cổ phiếu giảm trong ngày giao dịch không hưởng quyền tại thị trường chứng khoán Việt Nam chưa?

Let's detail the characteristics across major financial assets:

- **Cash:** does not generate any return unless invested. In the short term, excess cash is typically placed in interest-bearing bank accounts or short-term debt instruments.
- **Equities:** generate returns from capital appreciation and discretionary dividends (variable capital distributions). Rapidly growing companies often choose not to distribute dividends and instead reinvest earnings to fuel growth.
- **Bonds (Debt):** represent contractual lending agreements that pay fixed interest coupons (fixed capital distributions). Shorter-term bonds are dominated by capital distributions, whereas longer-term bonds exhibit substantial price volatility due to market interest rate movements.
- **Hybrids:** preferred stock and convertible bonds blend fixed-income features of debt with the growth potential of equity.

2.1.2 Types of Returns: Financial Instruments

Financial instruments are standardized legal agreements for financial assets (such as equity and bonds), as well as derivatives and securitized debt. The standardization facilitates public trading.

- **Cash Instruments:** represent direct ownership claims on financial assets (e.g., actual shares of stock or actual corporate bonds).
- **Derivatives:** financial instruments whose value is derived from the performance of another underlying financial asset or indicator (e.g., stock price, interest rate, index, exchange rate). A call option is a typical example.

2.1.3 Types of Returns: Financial Indicators

Financial indicators do not directly generate cash flows, but are critical to financial market operations.

- **Interest Rates:** traded through risk-free government bonds. Their level varies inversely with bond prices.
- **The Risk-Free Rate:** represented by short-term, risk-free government debt (e.g., US Treasury bills).

Exam Tip

Short-term discount instruments (like T-bills) do not pay regular coupons ($\text{Inc}_t = 0$). Instead, they sell at a discount to face value, and the return is earned purely from the price appreciation when the bond matures at par value.

Let's review the annualized risk-free rate calculation using a Treasury Bill:

$$r_{\text{period}} = \frac{P_1 - P_0 + \text{Inc}_t}{P_0} = \frac{\text{Face Value} - P_0}{P_0} \quad (2.4)$$

To annualize this period return for a T-bill with 90 days to maturity using 365 days in a year:

$$r_{\text{annual}} = (1 + r_{\text{period}})^{\frac{365}{90}} - 1 \quad (2.5)$$

Ví dụ: Một tín phiếu kho bạc Mỹ (T-Bill) thời hạn 90 ngày, mệnh giá 1,000 USD đang được bán ở mức giá 990 USD (tức chiết khấu 10 USD). Suất sinh lời thời kỳ là: $r_{\text{period}} = \frac{1000-990}{990} = 1.01\%$. Quy đổi ra suất sinh lời năm có ghép lãi (lấy 365 ngày): $r_{\text{annual}} = (1 + 0.0101)^{365/90} - 1 \approx 4.16\%$. Nếu chỉ tính lãi đơn bằng cách nhân 4 ($1.01\% \times 4 = 4.04\%$), bạn sẽ đánh giá thấp suất sinh lời thực tế nhận được. Hãy chú ý công thức ghép lãi này trong bài thi!

- **Foreign Exchange (FX) Rates:** represent the value of one currency expressed in terms of another currency.

When investing internationally, the home-currency return (r_{home}) depends on both the investment return in foreign currency (r_{foreign}) and the currency appreciation/depreciation rate (r_{FX}):

$$r_{\text{home}} = (1 + r_{\text{foreign}})(1 + r_{\text{FX}}) - 1 \approx r_{\text{foreign}} + r_{\text{FX}} \quad (2.6)$$

Where:

$$r_{\text{FX}} = \frac{\text{Ending FX Rate} - \text{Beginning FX Rate}}{\text{Beginning FX Rate}} \quad (2.7)$$

Ví dụ: Một nhà đầu tư USD mua trái phiếu bằng EUR. Suất sinh lời bằng EUR là $r_{\text{foreign}} = -2.91\%$. Cùng năm đó, tỷ giá EUR/USD biến động khiến đồng EUR giảm giá so với USD là -6.90% (tức $r_{\text{FX}} = -6.90\%$, tỷ giá thay đổi từ 1.2216 USD/EUR xuống 1.1373 USD/EUR). Suất sinh lời cuối cùng bằng USD của nhà đầu tư này sẽ chịu tác động kép của cả hai khoản lỗ: $r_{\text{home}} = (1 - 0.0291)(1 - 0.0690) - 1 = -9.61\%$. Việc đồng ngoại tệ mất giá đã khuếch đại khoản lỗ của nhà đầu tư khi quy đổi về nội tệ.

- **Market Indexes:** reflect the aggregate value of a basket of assets (e.g., S&P 500, OMX Copenhagen 25 Index) to indicate market trends.

To evaluate and compare performance, we look at the arithmetic and geometric averages of index returns:

$$\bar{r}_{\text{arithmetic}} = \frac{1}{T} \sum_{t=1}^T r_t \quad (2.8)$$

$$\bar{r}_{\text{geometric}} = \left[\prod_{t=1}^T (1 + r_t) \right]^{\frac{1}{T}} - 1 \quad (2.9)$$

Let's review the historical performance numbers of the OMX Copenhagen 25 Index versus Ørsted A/S equity between 2017 and 2023:

Table 2.1: OMX Copenhagen 25 Index vs. Ørsted A/S Equity Total Returns (2017–2023)

Year	OMX Copenhagen 25 Return	Ørsted A/S Equity Return
2017	15.09%	28.81%
2018	-11.19%	31.30%
2019	28.71%	60.37%
2020	35.60%	82.00%
2021	18.60%	-31.91%
2022	-11.54%	-22.92%
2023	9.63%	-38.57%
Arithmetic Mean	12.13%	15.58%
Geometric Mean	10.81%	6.86%

Nhìn vào bảng số liệu trên, bạn thấy điều gì thú vị? Suất sinh lời trung bình số học (Arithmetic Mean) của Ørsted (15.58%) cao hơn của OMX index (12.13%), nhưng suất sinh lời trung bình hình học (Geometric Mean) của nó lại thấp hơn rất nhiều (6.86% so với 10.81%). Điều này chứng minh tác động tiêu cực của biến động (volatility drag) lên lãi kép: Ørsted có những năm tăng trưởng rất nóng (82.00%) nhưng sau đó lại sụt giảm rất sâu (-38.57%), làm xói mòn hiệu quả tăng trưởng vốn tích lũy của nhà đầu tư dài hạn.

2.2 LONG-TERM FINANCIAL ASSET RETURNS

- calculate, compare, and interpret different types of returns for financial assets, instruments, and indicators

Lịch sử thị trường tài chính cung cấp những bài học vô giá về hành vi dài hạn của các lớp tài sản khác nhau. Trong phần này, chúng ta sẽ rà soát số liệu thực tế qua 45 năm (1979–2023) tại thị trường Mỹ để so sánh lợi nhuận giữa các loại tài sản và hiểu rõ cấu phần sinh lời của chúng thay đổi thế nào qua thời gian.

2.2.1 Historical Long-Term Asset Returns

Based on historical data for selected major US asset classes between 1979 and 2023 (large-cap stocks, small-cap stocks, corporate bonds, government bonds, and Treasury bills), several trends emerge:

- **Equities Outperform Bonds:** driven primarily by capital appreciation, equities deliver higher long-term average returns than fixed-income indexes.
- **Small-Cap Premium:** small-capitalization stocks have periodically outperformed large-capitalization stocks (size premium), though this premium varies over time and can occasionally be negative.
- **Corporate vs. Government Debt:** corporate bonds offer higher coupon yields on paper, but government bonds can occasionally match or exceed corporate bond performance over specific horizons due to the impact of default losses.

Let's review the relative outperformance calculations using geometric average annual returns from the 1979–2023 historical record:

- Large-cap equity (12.26%) versus corporate debt (7.43%):

$$\text{Outperformance} = \frac{1 + 0.1226}{1 + 0.0743} - 1 = 4.50\% \quad (2.10)$$

- Small-cap equity (11.23%) versus large-cap equity (12.26%):

$$\text{Small-Cap Premium} = \frac{1 + 0.1123}{1 + 0.1226} - 1 = -0.92\% \quad (\text{a discount over this specific period}) \quad (2.11)$$

- US Treasury bonds (7.62%) versus corporate bonds (7.43%):

$$\text{Outperformance} = \frac{1 + 0.0762}{1 + 0.0743} - 1 = 0.18\% \quad (2.12)$$

2.2.2 Long-Term Asset Return Components

The relative contribution of capital appreciation and capital distributions has evolved significantly:

- **Equities:** dividend yields have trended downward over time, particularly following a 1982 regulatory change that established a safe harbor for share buybacks. Companies increasingly repurchase shares to return cash to shareholders, which reduces share count, increases stock value, and provides tax advantages (capital gains tax is generally lower than dividend tax).
- **Government Bonds:** income return (coupon payments) remains the dominant driver of total returns, historically accounting for approximately 85% of total government bond returns. Capital appreciation plays a minor role, and bond prices suffer capital losses during periods of rising interest rates (such as the early 2020s).

Hãy ghi nhớ: Mua lại cổ phiếu (Share Buybacks) không làm phát sinh dòng tiền phân phối trực tiếp cho cổ đông như cổ tức tiền mặt, do đó nó được hạch toán vào cấu phần tăng giá (Price Return). Đây là lý do tại sao phần lợi nhuận tăng giá của cổ phiếu lại chiếm tỷ trọng lớn dần trong những thập kỷ gần đây tại các nước phát triển. Bạn có nghĩ đây là xu hướng tất yếu của các doanh nghiệp khi tối ưu hóa thuế cho cổ đông không?

2.3 RISKS AND RETURNS

- | | |
|---|--|
| a | calculate, compare, and interpret different types of returns for financial assets, instruments, and indicators |
|---|--|

Rủi ro tài chính là khả năng suất sinh lời thực tế nhận được lệch khỏi suất sinh lời kỳ vọng. Việc đo lường rủi ro và xây dựng danh mục đầu tư đòi hỏi chúng ta phải phân loại rủi ro thành hệ thống (Systematic Risk) và phi hệ thống (Unsystematic Risk), đồng thời áp dụng nguyên lý đánh đổi rủi ro - lợi nhuận (Risk-Return Trade-Off). Cùng rà soát các định nghĩa và cách tính suất sinh lời danh mục ở phần dưới đây.

Risk represents the uncertainty of investment returns and can be categorized into macro and micro risks:

- **Systematic Risk (Macro Risks):** broad, economy-wide risks that affect all financial assets (e.g., inflation risk, market interest rate risk, and market risk). Systematic risk cannot be diversified away.
- **Unsystematic Risk (Micro Risks):** risks specific to a particular issuer or sector (e.g., credit/default risk, business risk, financial leverage, sector/industry risk). Unsystematic risk can be mitigated through diversification.

2.3.1 The Risk–Return Trade-Off

Investors demand a higher expected return (risk premium) to compensate for taking on higher volatility and uncertainty. Exhibit 2.1 shows that historically, higher average annual returns are generally associated with a wider range of annual returns (higher volatility).

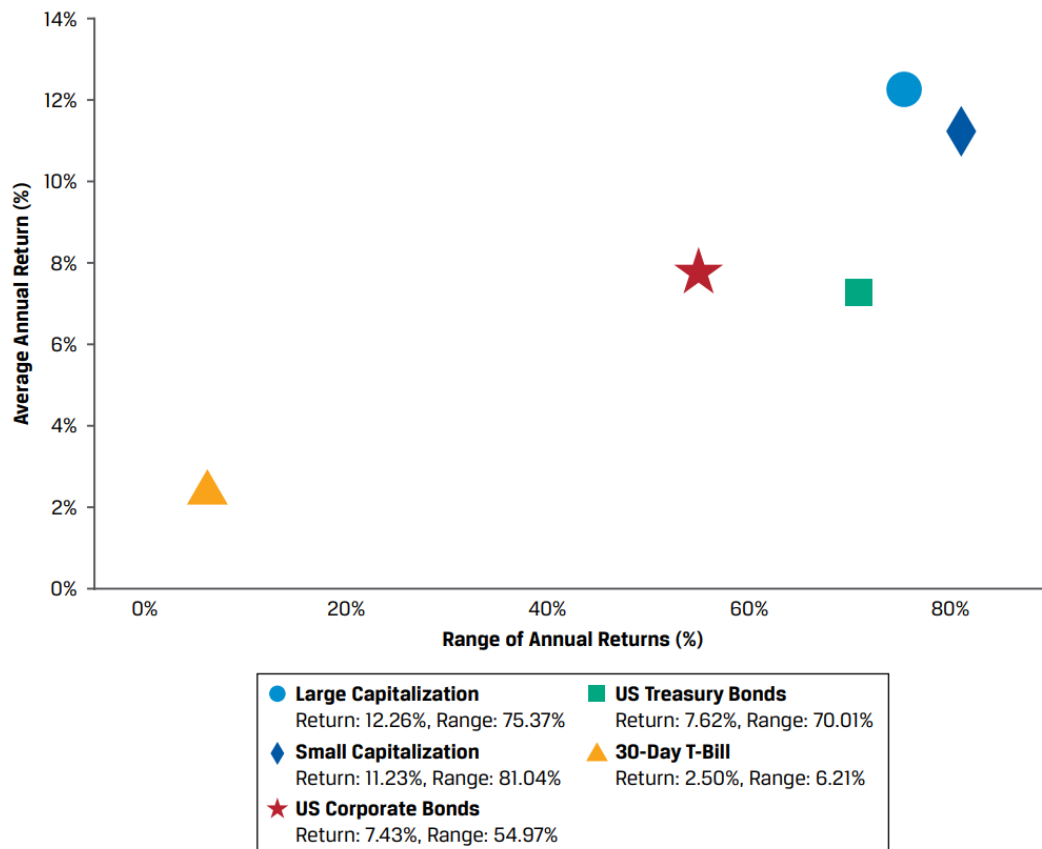


Figure 2.1: Historical Geometric Average Return for Different Assets (1979–2023)

2.3.2 Portfolio Return Calculation

The expected return of a portfolio is the weighted average of the returns of the individual assets in the portfolio:

$$r_p = \sum_{i=1}^N w_i r_i = w_1 r_1 + w_2 r_2 + \cdots + w_N r_N \quad (2.13)$$

Where:

w_i is the weight of asset i in the portfolio (where $\sum_{i=1}^N w_i = 1$).

r_i is the rate of return of asset i .

Let's review the asset allocations and long-term returns for three different investors (Investor A: Aggressive, Investor B: Balanced, and Investor C: Conservative):

Table 2.2: Asset Class Returns and Allocation Summary

Asset Class	Total Return	Capital Appreciation	Capital Distribution
Large-cap Equities	12.00%	8.50%	3.50%
Long-term Gov Bonds	5.75%	0.75%	5.00%
30-Day T-Bills (Cash)	3.00%	0.00%	3.00%

Table 2.3: Investor Asset Allocations and Portfolio Returns

Investor	Equities Weight	Bonds Weight	T-Bills Weight	Total Return	Capital Apprec.	Capital Distrib.
A	80%	10%	10%	10.48%	6.88%	3.60%
B	60%	35%	5%	9.36%	5.36%	4.00%
C	50%	45%	5%	8.74%	4.59%	4.15%

Ví dụ tính chi tiết cho danh mục Investor A: Với cấu trúc phân bổ 80% cổ phiếu, 10% trái phiếu chính phủ dài hạn, và 10% tín phiếu kho bạc (tiền mặt): - Tỷ suất sinh lời tổng thể: $r_p = 80\% \times 12.00\% + 10\% \times 5.75\% + 10\% \times 3.00\% = 9.60\% + 0.58\% + 0.30\% = 10.48\%$. - Tỷ suất sinh lời từ tăng trưởng giá: $r_{price} = 80\% \times 8.50\% + 10\% \times 0.75\% + 10\% \times 0.00\% = 6.80\% + 0.08\% + 0.00\% = 6.88\%$. - Tỷ suất sinh lời từ phân phối thu nhập: $r_{dist} = 80\% \times 3.50\% + 10\% \times 5.00\% + 10\% \times 3.00\% = 2.80\% + 0.50\% + 0.30\% = 3.60\%$. Như vậy, danh mục của Investor A được thiết kế để tối đa hóa phần lãi vốn (Capital Appreciation), trong khi Investor C ưu tiên dòng thu nhập đều đặn (Capital Distribution). Bạn hãy thử bấm máy tính tự giải lại trường hợp của Investor B và C để thành thạo kỹ năng tính toán này nhé!

Module 3

Benchmarking Returns

LEARNING OUTCOMES

a	calculate and compare money-weighted and time-weighted rates of return
b	describe the choices and the implications of the different weighting methods used in index construction and management, and calculate, interpret, and explain the value and the returns of an index

Money-Weighted and Time-Weighted Rates of Return

Index Definition and Calculation

Price Return and Total Return Indexes

Index Weightings

Index Management

Comparing Index Performance

Module 4

The Time Value of Money in Finance

LEARNING OUTCOMES

a	calculate and interpret the present value of fixed-income and equity instruments based on expected future cash flows
b	calculate and interpret the implied return of fixed-income instruments and required return and implied growth of equity instruments given their present value and cash flows
c	explain the cash flow additivity principle and its importance for the condition of no arbitrage, and explain its use in calculating implied forward interest rates, forward exchange rates, and option values

Time Value of Money in Fixed Income and Equity

Fixed-Income Instruments

Equity Instruments

Implied Return and Growth

Implied Return for Fixed-Income Instruments

Equity Instruments, Implied Return, and Implied Growth

Cash Flow Additivity

Implied Forward Rates Using Cash Flow Additivity

Forward Exchange Rates Using No Arbitrage

Option Pricing Using Cash Flow Additivity

Module 5

Statistical Characteristics of Asset Returns

LEARNING OUTCOMES

a	calculate, interpret, and evaluate various measures of (1) central tendency and location and (2) dispersion
b	describe, interpret, and evaluate measures of skewness and kurtosis
c	calculate, interpret, and evaluate covariance and correlation
d	calculate, interpret, and evaluate semi-deviation and coefficient of variation

Measures of Central Tendency and Dispersion

Measures of Central Tendency

Measures of Dispersion

Measures of Shape of a Distribution

Skewness

Kurtosis

Covariance and Correlation between Variables

Covariance

Correlation

Other Dispersion Measures

Semi-Deviation

Coefficient of Variation

Module 6

Statistical Distributions for Financial Asset Prices and Returns

LEARNING OUTCOMES

a	calculate, interpret, and evaluate unconditional expected values for mean, variance, and covariance
b	calculate, interpret, and evaluate the principal moments of key statistical distributions used in finance
c	calculate, interpret, and evaluate conditional expectations, variances, and covariances
d	formulate investment problems through Bayesian updating

Random Variables and Unconditional Expectations

Random Variables and Their Properties

Unconditional Expectations, Variance, and Covariance

Key Statistical Distributions in Finance

Discrete Statistical Distributions

Continuous Statistical Distributions

Practical Implementation

Conditional Expectations, Variances, and Covariances

Joint Distributions of Random Variables

Conditional Expectations

Conditional Means, Variances, and Covariances

Bayesian Updating

Module 7

Estimation and Hypothesis Testing

LEARNING OUTCOMES

a	explain the central limit theorem and the application of confidence intervals and sampling methodologies
b	explain hypothesis testing and its components, including statistical significance, Type I and Type II errors, and the power of a test; construct appropriate hypothesis tests; and interpret the results
c	compare and contrast parametric and non-parametric tests, describe situations in which each is the more appropriate type of test, construct appropriate hypothesis tests, and interpret the results

The Central Limit Theorem, Confidence Intervals, and Sampling

The Central Limit Theorem

Confidence Intervals

Sampling

Hypothesis Testing and Parametric Testing

The Hypothesis Testing Process

Parametric Tests for a Single Variable

Parametric Tests for Multiple Variables

Tests of the Differences in Variances

Parametric Test of Correlations

P-Hacking

Practical Implementation

Non-Parametric Tests

Non-Parametric Test for Correlation

Tests of Independence

Module 8

The Return and Risk of a Financial Portfolio

LEARNING OUTCOMES

a	calculate, interpret, and evaluate the expected return, variance, standard deviation, covariance, and correlation of portfolio returns
b	describe, calculate, and interpret the minimum-variance portfolio and portfolios that lie on the efficient frontier
c	explain the selection of an optimal portfolio, given an investor's risk aversion and the capital allocation line, and how this extends to the market portfolio and the capital market line

Calculating Portfolio Statistics

Returns of a Portfolio

Variance and Standard Deviation of Portfolio Returns

The Benefits of Diversification

Types of Returns: Financial Indicators

Long-Term Financial Asset Returns

Risks and Returns

The Minimum-Variance Portfolio and the Efficient Frontier

The Minimum-Variance Portfolio

The Minimum-Variance Frontier

Portfolio Optimization

Optimal Portfolio Selection

The Capital Allocation Line (CAL)

The Capital Market Line (CML)

The Capital Asset Pricing Model (CAPM)

Limitations

Module 9

Simulation of Financial Asset Prices and Returns

LEARNING OUTCOMES

a	describe historical simulation and explain how it can be used in investment applications
b	describe bootstrap resampling, and explain how it can be used in investment applications
c	describe Monte Carlo simulation and explain how it can be used in investment applications

Historical Simulation

- Overview of Simulation Methods
- Historical Simulation Overview
- Implementing Historical Simulation
- Features of Historical Simulation

Bootstrapping

- Bootstrapping Overview
- Implementing Bootstrapping
- Features of Bootstrapping Simulations

Monte Carlo Simulation

- Monte Carlo Simulation Overview
- Multivariate Monte Carlo Simulation
- Features of Monte Carlo Simulation
- Practical Implementation of Simulation Approaches Using Excel and Other Tools

Module 10

Applications of Simple Linear Regression in Finance

LEARNING OUTCOMES

a	describe, interpret, and explain simple linear regression, including coefficient estimation using the least squares criterion
b	describe and compare the assumptions of simple linear regression, identify violations through analyzing residuals, evaluate the estimated model's goodness-of-fit and regression coefficients, and results of ANOVA estimates
c	calculate and interpret predicted values, the standard error of the estimate, and prediction intervals for the dependent variable in a simple linear regression model and describe different functional forms
d	calculate and interpret the variable estimates of the capital asset pricing model (CAPM)

Linear Regression Estimation

- Linear Regression Analysis
- Coefficient Estimation of Linear Regression
- Estimation Residuals
- Data Types

Limitations of Linear Regression Models

- Assumptions of the Simple Linear Regression Model
- Residual Analysis
- Evaluating Goodness-of-Fit and Regression Coefficients
- Hypothesis Testing

Using Linear Regression Models for Prediction

- Prediction Using a Simple Regression
- Modeling Non-Linear Relationships in Economic and Financial Data
- Practical Implementation
- Hypothesis Testing

Using Linear Regression Models to Evaluate Financial Assets: The Capital Asset Pricing Model

Module 11

Introduction to Financial Data Science

LEARNING OUTCOMES

- | | |
|---|---|
| a | describe how big data, machine learning, and artificial intelligence are used in financial data science, fintech, and investment management |
|---|---|
-

Introduction to Financial Data Science

Financial Data

Big Data

Financial Data Science

Advanced Analytical Tools: AI and ML

Tackling Big Data with Data Science

List of Tables

2.1	OMX Copenhagen 25 Index vs. Ørsted A/S Equity Total Returns (2017–2023)	13
2.2	Asset Class Returns and Allocation Summary	15
2.3	Investor Asset Allocations and Portfolio Returns	16

List of Figures

1.1	The Impact of Compounding	7
1.2	Components of Required Rate of Return and Examples of Different Risk Premia	8
1.3	The Relationship between Total Return, Gross Return, and Net Return	9
1.4	The Payoff Relationship between Stock Price, Unleveraged Return, and Leveraged Return	10
2.1	Historical Geometric Average Return for Different Assets (1979–2023)	15

putModule/Module10.tex

List of Tables

List of Figures